MATLAB/Simulink Modeling

Mathematical equations for the motors:

Electrical equation:

$$\boxed{\frac{di(t)}{dt} = \frac{V(t) - Ri(t) - K_e\omega(t)}{L}}$$

Electromagnetic torque:

$$\boxed{T_m(t) = K_t i(t)}$$

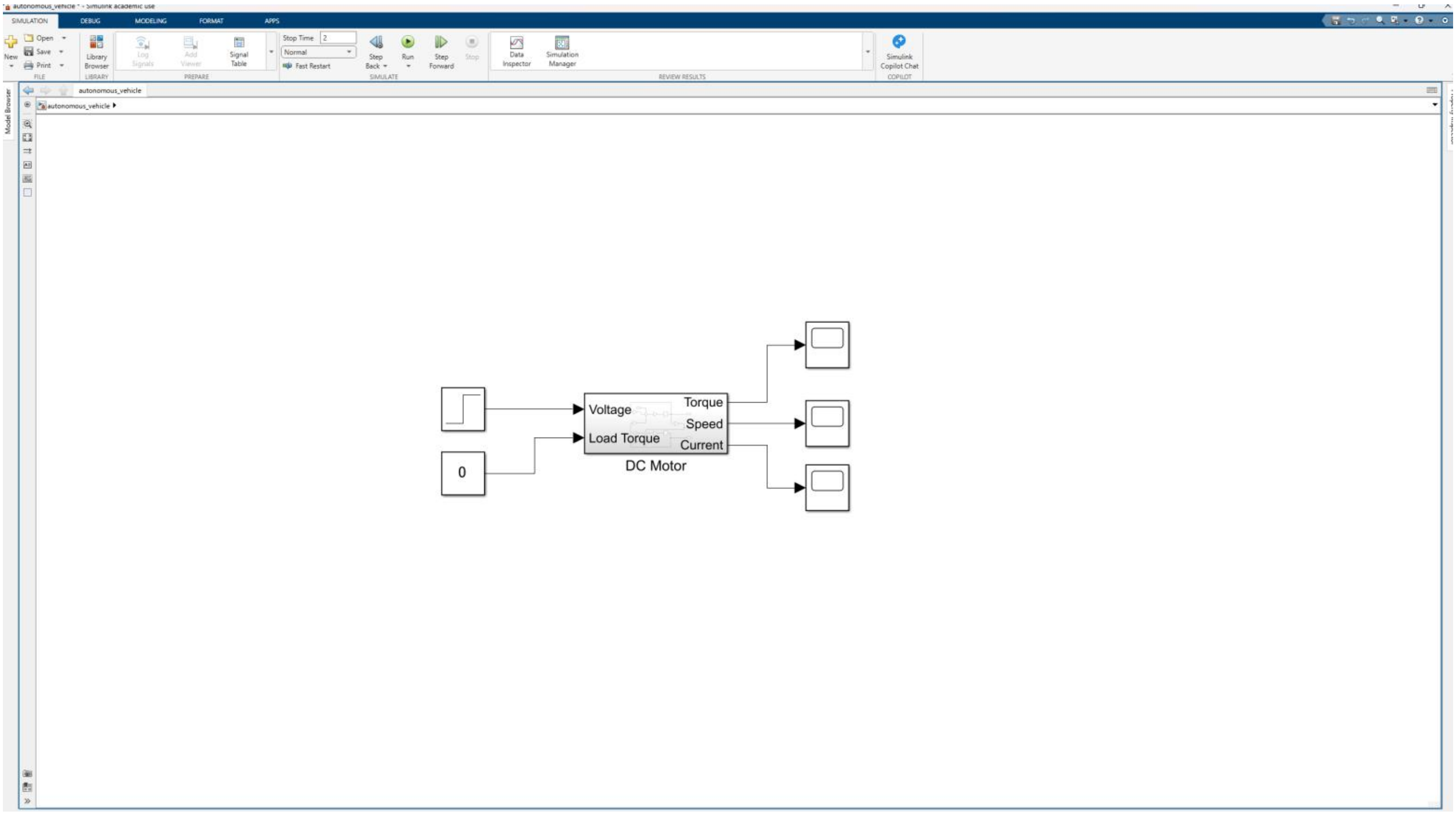
Mechanical equation:

$$\boxed{\frac{d\omega(t)}{dt} = \frac{K_t i(t) - b\omega(t) - T_L(t)}{J}}$$

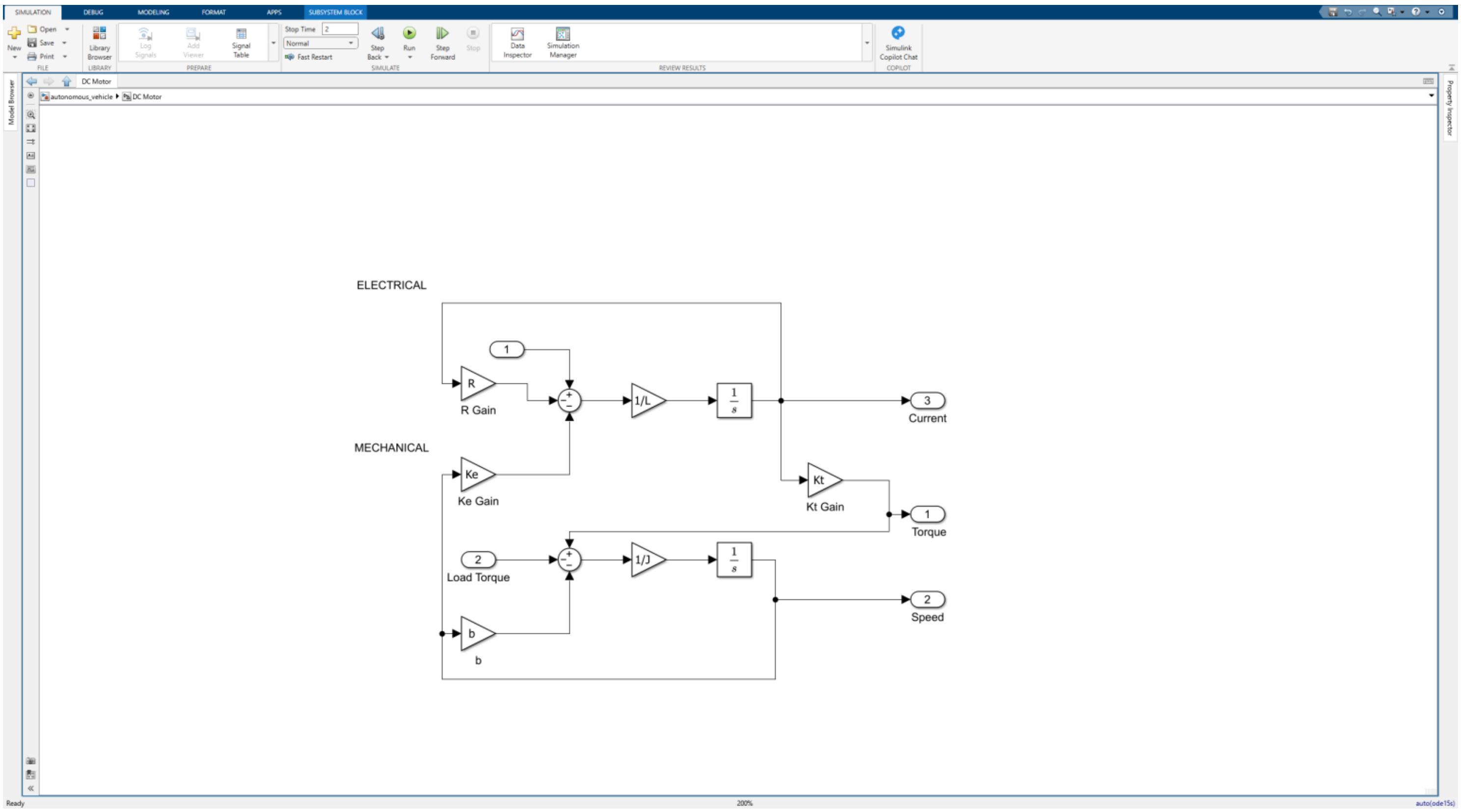
Transfer function from voltage to motor speed:

$$\boxed{\frac{\Omega(s)}{V(s)} = \frac{K_t}{LJs^2 + (Lb + RJ)s + (Rb + K_e K_t)}}$$

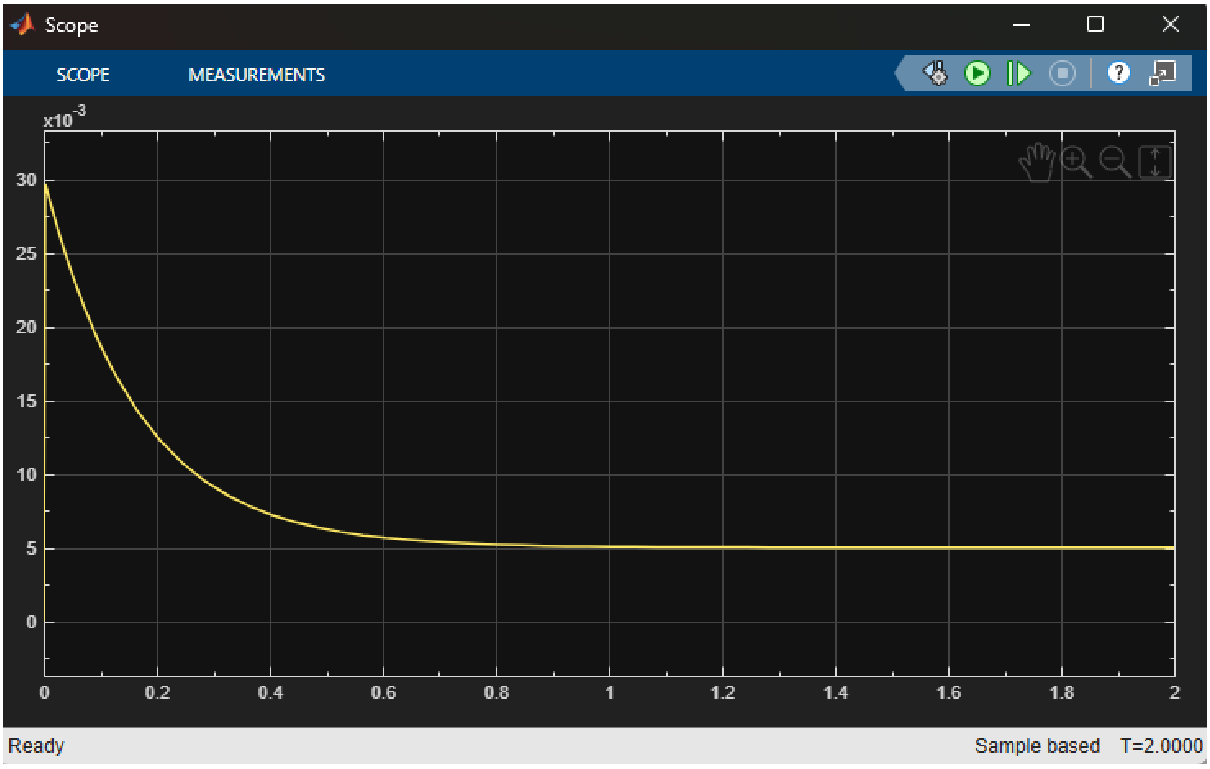
Simulink model:



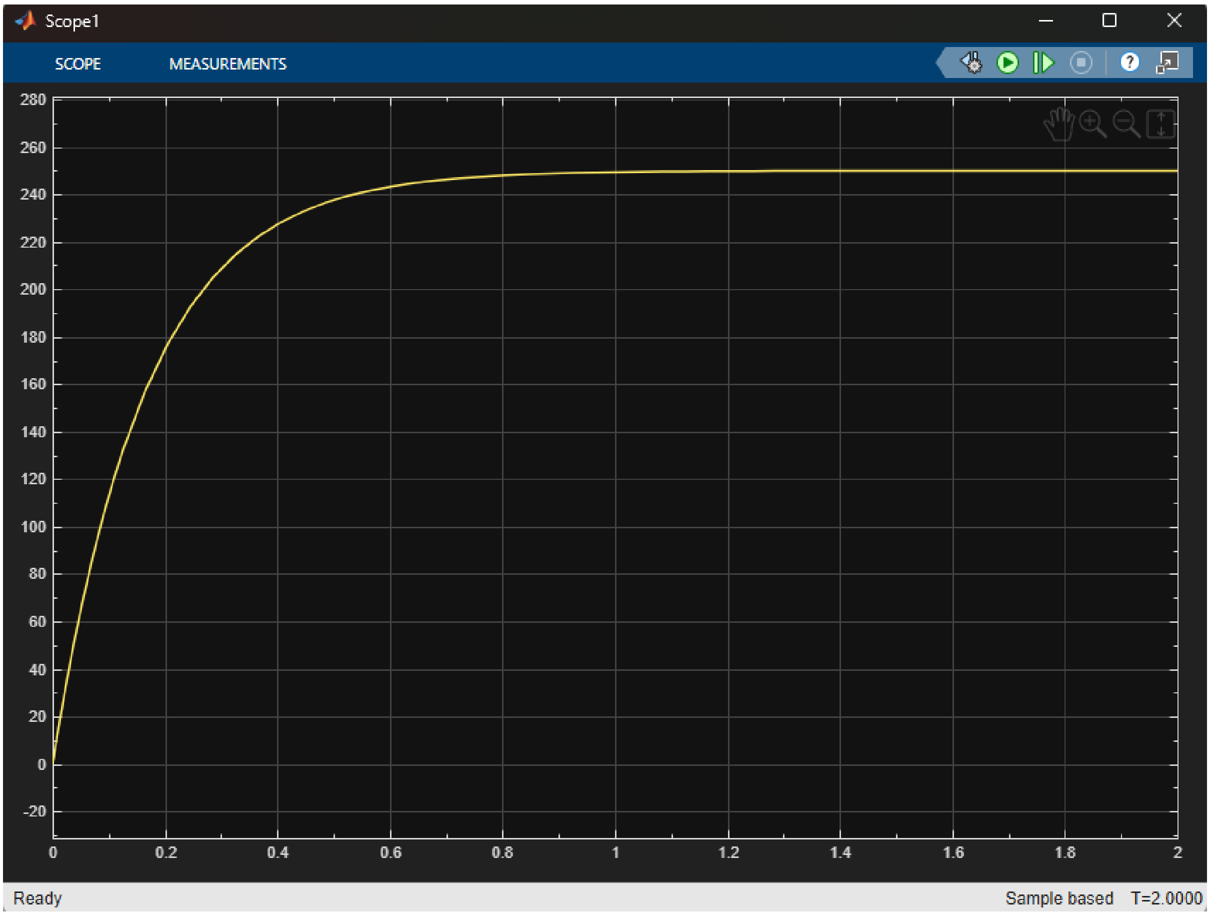
Subsystem model:



Torque Behaviour:



Speed Behaviour:



Current Behaviour: